



12429 - Testing the formation of a trail of dwarf galaxies through tip of the red giant branch distances

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	DF9-RCP17	NIRCam Imaging	(1) DF9
	7	NGC1035-DF4	NIRCam Imaging	(7) NGC1035
	2	DF4&DF5-LEDA	NIRCam Imaging	(2) DF4+DF5
	3	RCP16	NIRCam Imaging	(3) RCP16
	5	DF7	NIRCam Imaging	(5) DF7
	6	Stream1-N1052halo	NIRCam Imaging	(6) Stream1

ABSTRACT

Although dark matter is understood as providing the basic scaffolding for galaxy formation, there are dwarf galaxies in the NGC 1052 group with highly unusual properties that suggest an alternative origin. Two of these dwarfs (DF2 and DF4) are known to be deficient in dark matter and to host unusually large, luminous globular clusters with homogeneous colors. They are also part of an apparent linear structure or "trail" of ~ 10 dwarfs, which may be explained as the residue of an ancient "bullet dwarf" event that could also naturally explain the anomalous dark matter and globular cluster properties. We propose a definitive test of this scenario, using NIRCам imaging of dwarfs in the group to measure the distances to the dwarfs using the tip of the red giant branch, testing for a linear dependence on sky position. Confirmation of the trail, and mapping of its full three-dimensional structure, will provide important input to novel mechanisms for galaxy formation and to constraints on the nature of dark matter.

OBSERVING DESCRIPTION

This program studies the low surface brightness dwarf galaxies in the NGC 1052 group using NIRCам, aiming for measuring their distance through the tip of the red giant branch (TRGB), a standard distance indicator that has been applied to the dark-matter free galaxies NGC 1052-DF2 and DF4 using HST.

We will target 5 dwarfs in the group with 4 pointings, with the filter combination F090W in SWC and F277W in LWC. The TRGB magnitude will be measured in F090W, and the F090W-F277W color information will be used to determine the stellar population for TRGB calibration, and also remove the red background contamination. Two additional pointings are used for the stellar stream of the group host NGC 1052, and for the stellar halo of the NGC 1035.

4 parallel NIRISS pointings will be used when there is secondary target in NIRISS field, mainly for the diffuse star light studies, with bonus measurements attempted for the TRGB.

Proposal 12429 - Targets - Testing the formation of a trail of dwarf galaxies through tip of the red giant branch distances

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	DF9	RA: 02 40 7.0241 (40.0292671d) Dec: -08 13 44.47 (-8.22902d) Equinox: J2000 Comments: Category=Galaxy Description=[Dwarf galaxies]		
(2)	DF4+DF5	RA: 02 39 15.0503 (39.8127096d) Dec: -08 06 57.50 (-8.11597d) Equinox: J2000 Comments: Category=Galaxy Description=[Dwarf galaxies]		
(3)	RCP16	RA: 02 39 39.3240 (39.9138500d) Dec: -08 13 42.42 (-8.22845d) Equinox: J2000 Comments: Category=Galaxy Description=[Dwarf galaxies]		
(4)	LEDA4014647	RA: 02 38 48.4944 (39.7020600d) Dec: -08 02 57.77 (-8.04938d) Equinox: J2000 Comments: Category=Galaxy Description=[Dwarf galaxies]		
(5)	DF7	RA: 02 38 29.7624 (39.6240100d) Dec: -07 56 29.74 (-7.94159d) Equinox: J2000 Comments: The actual coordinate of DF7 is 02 38 29.7624, -07 55 32.74. If we use this, then there will be too much charged time to include the full module for RCP 16. For unknown reasons, the charged time is less if we use -07 56 29.74 as the coordinate and then do a manual offset back to the target. Category=Galaxy Description=[Dwarf galaxies]		
(6)	Stream1	RA: 02 41 28.9900 (40.3707917d) Dec: -08 23 4.92 (-8.38470d) Equinox: J2000 Comments: Category=Galaxy Description=[Dwarf galaxies]		
(7)	NGC1035	RA: 02 39 30.7600 (39.8781667d) Dec: -08 07 57.00 (-8.13250d) Equinox: J2000 Comments: Category=Galaxy Description=[Spiral galaxies]		

Proposal 12429 - Observation 1 - Testing the formation of a trail of dwarf galaxies through tip of the red giant branch distances

Observation	Proposal 12429, Observation 1: DF9-RCP17										Thu Jun 18 21:00:09 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCam Imaging											
	Coordinated Parallel Template(s): NIRISS Imaging											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	DF9	RA: 02 40 7.0241 (40.0292671d) Dec: -08 13 44.47 (-8.22902d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Dwarf galaxies]											
Template	NIRCam Imaging					NIRISS Imaging						
	Module: B											
	Subarray: FULL											
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes
	1	INTRAMODULEBOX		4				1		NIRCam Only		NO_DITHERING
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID		
	1	F090W	F277W	SHALLOW4	10	1	4	4	2104.407			
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	F200W		NIS	12	1	4	4	2104.407			

Proposal 12429 - Observation 1 - Testing the formation of a trail of dwarf galaxies through tip of the red giant branch distances

Special Requirements	Aperture PA Range 51.05262691 to 71.05262691 Degrees (V3 51.0 to 71.0) Offset 40.0 arcsec, -30.0 arcsec No Parallel Attachments
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Proposal 12429 - Observation 7 - Testing the formation of a trail of dwarf galaxies through tip of the red giant branch distances

Observation	Proposal 12429, Observation 7: NGC1035-DF4										Thu Jun 18 21:00:09 GMT 2026		
	Diagnostic Status: Warning												
	Observing Template: NIRCcam Imaging												
	Coordinated Parallel Template(s): NIRISS Imaging												
Diagnos	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(7)	NGC1035	RA: 02 39 30.7600 (39.8781667d) Dec: -08 07 57.00 (-8.13250d) Equinox: J2000										
	Comments: Category=Galaxy Description=[Spiral galaxies]												
Template	NIRCcam Imaging					NIRISS Imaging							
	Module: ALL												
	Subarray: FULL												
	Target Placement: Module B center (small extended source)												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes	
	1	INTRAMODULEBOX		4				1		NIRCcam Only		NO_DITHERING	
Spectral Elements	NIRCcam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID			
	1	F090W	F277W	SHALLOW4	10	1	4	4	2104.407				
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	F200W		NIS	12	1	4	4	2104.407				

Proposal 12429 - Observation 7 - Testing the formation of a trail of dwarf galaxies through tip of the red giant branch distances

Special Requirements	Aperture PA Range 51.05262691 to 71.05262691 Degrees (V3 51.0 to 71.0) Offset 0.0 arcsec, 25.0 arcsec No Parallel Attachments Fiducial Point Override NRCBS_FULL
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Proposal 12429 - Observation 2 - Testing the formation of a trail of dwarf galaxies through tip of the red giant branch distances

Observation	Proposal 12429, Observation 2: DF4&DF5-LEDA										Thu Jun 18 21:00:09 GMT 2026		
	Diagnostic Status: Warning												
	Observing Template: NIRCam Imaging												
	Coordinated Parallel Template(s): NIRISS Imaging												
Diagnostics	(DF4&DF5-LEDA (Obs 2)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.												
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(2)	DF4+DF5	RA: 02 39 15.0503 (39.8127096d) Dec: -08 06 57.50 (-8.11597d) Equinox: J2000										
	Comments: Category=Galaxy Description=[Dwarf galaxies]												
Template	NIRCam Imaging					NIRISS Imaging							
	Module: ALL												
	Subarray: FULL												
	Target Placement: Module gap (large extended source)												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes	
	1	INTRAMODULEBOX		4				1		NIRCam Only		NO_DITHERING	
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID			
	1	F090W	F277W	SHALLOW4	10	1	4	4	2104.407				
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	F200W		NIS	12	1	4	4	2104.407				

Proposal 12429 - Observation 2 - Testing the formation of a trail of dwarf galaxies through tip of the red giant branch distances

Special Requirements	Aperture PA Range 50.92542306 to 70.92542306 Degrees (V3 51.0 to 71.0) Offset -60.0 arcsec, 35.0 arcsec No Parallel Attachments
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Proposal 12429 - Observation 3 - Testing the formation of a trail of dwarf galaxies through tip of the red giant branch distances

Observation	Proposal 12429, Observation 3: RCP16										Thu Jun 18 21:00:09 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRCam Imaging										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	RCP16	RA: 02 39 39.3240 (39.9138500d) Dec: -08 13 42.42 (-8.22845d) Equinox: J2000								
Template	Comments: Category=Galaxy Description=[Dwarf galaxies]										
	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F277W	MEDIUM2	8	1	4	4	3092.19		
Special Requirements	Fiducial Point Override NRCB1_FULL										

Proposal 12429 - Observation 5 - Testing the formation of a trail of dwarf galaxies through tip of the red giant branch distances

Observation	Proposal 12429, Observation 5: DF7										Thu Jun 18 21:00:09 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCam Imaging											
	Comments: We don't know why the actual coordinate of DF7 (39.62401, -7.92576) cannot make it schuduled here. Therefore we use a different coordinate, and make use of the big FoV to catch DF7.											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(5)	DF7	RA: 02 38 29.7624 (39.6240100d) Dec: -07 56 29.74 (-7.94159d) Equinox: J2000									
	Comments: The actual coordinate of DF7 is 02 38 29.7624, -07 55 32.74. If we use this, then there will be too much charged time to include the full module for RCP 16. For unknown reasons, the charged time is less if we use -07 56 29.74 as the coordinate and then do a manual offset back to the target.											
	Category=Galaxy Description=[Dwarf galaxies]											
Template	Module						Subarray					
	B						FULL					
Dithers	#	Primary Dither Type			Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX			4		STANDARD				1	
Spectral Elements	#	Short Filter		Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W		F277W	MEDIUM2	8	1	4	4	3092.19		
Special Requirements	Offset 39.0 arcsec, -18.7 arcsec											

Proposal 12429 - Observation 6 - Testing the formation of a trail of dwarf galaxies through tip of the red giant branch distances

Observation	Proposal 12429, Observation 6: Stream1-N1052halo										Thu Jun 18 21:00:09 GMT 2026		
	Diagnostic Status: Warning												
	Observing Template: NIRCам Imaging												
	Coordinated Parallel Template(s): NIRISS Imaging												
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(6)	Stream1	RA: 02 41 28.9900 (40.3707917d) Dec: -08 23 4.92 (-8.38470d) Equinox: J2000										
	Comments: Category=Galaxy Description=[Dwarf qalaxies]												
Template	NIRCам Imaging					NIRISS Imaging							
	Module: ALL												
	Subarray: FULL												
	Target Placement: Module A center (small extended source)												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes	
	1	INTRAMODULEBOX		4				1		NIRCам Only		NO_DITHERING	
Spectral Elements	NIRCам Imaging		Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID		
	1	F090W		F277W	SHALLOW4	10	1	4	4	2104.407			
Spectral Elements	NIRISS Imaging		Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	F090W			NIS	12	1	4	4	2104.407			

Proposal 12429 - Observation 6 - Testing the formation of a trail of dwarf galaxies through tip of the red giant branch distances

Special Requirements	Aperture PA Range 50.86565232 to 70.86565232 Degrees (V3 51.0 to 71.0) Offset 0.0 arcsec, 35.0 arcsec No Parallel Attachments Fiducial Point Override NRCAS_FULL
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